SARASWAT ACADEMY

TEST- PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

(CLASS-X)

Q1: For what value k, do the equations $2x - y + 3 = 0$ and $6x - ky + 9 = 0$ represent coincident lines?

(a) 2 (b) – 2 (c) 3 (d) -3

Q2: What are the values of x and y for the following pair of linear equations?

$99x + 101y = 499$ and $101x + 99y = 501$

(a) 3 and 6 (b) 3 and 2 (c) 2 and 3 (d) 6 and 3

Q3: The 2 digit number which becomes 5/6th of itself when its digits are reversed. The difference in the

digits of the number being 1, then the two digits number is-

(a) 45 (b) 54 (c) 36 (d) None of these

Q4: What do you say about the lines represented by: $2x + 3y - 9 = 0$ and $4x + 6y - 18 = 0$

(a) lines are parallel (b) lines are coincident

(c) lines are intersecting (d) can't say anything

<u>COMPETENCY BASED QUESTIONS</u>

Q1: Aftab tells his daughter, 7 years ago, I was seven times as old as you were then. Also, 3 years from now, I shall be three times as old as you will be. Find the present age of Aftab and his daughter.

Q2: A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction.

Q3: Aruna has only Rs 1 and Rs 2 coins with her. If the total number of coins that she has is 50 and the amount of money with her is Rs 75. then find the number of Rs 1 and Rs 2 coins.

Q4: Solve below simultaneous equations for x and y,

$$3x - 5y = 4 \text{ and } 9x - 2y = 7$$

Q5: The age of father is 22 years more than his son. In three years, the father's age will be twice that of his son. Find the present age of his son.

Q6: There are some students in two examination halls A and B. To make the number of students equal in each hall, 10 students are sent from A to B. But, if 20 students are sent from B to A, the number of students in A becomes double the number of students in B. Find the number of students in the two halls.